

Chapter One

General Pharmacology

Learning Objectives

- Up on completion of this course, students will be able to
 - Understand the general principles of pharmacology.
 - Explain the pharmacokinetics of drugs affecting the nervous system, respiratory system, gastrointestinal system and autacoids.
 - Explain the pharmacological actions, mechanism of actions, and therapeutic uses and pharmacokinetics of drugs affecting the nervous system, respiratory system, gastrointestinal system and autacoids.
 - Explain side effects of drugs acting on autonomic nervous system, central nervous system, respiratory, GI, and drugs acting on inflammation.
 - Apply concept and principles of pharmacology to ensure and proper use of drugs.
 - To work in Pharmacology Laboratory and will be able to practice selected basic experimental demonstration

Introduction

- ❖ Pharmacology is the science of how **drugs** act on biological systems and how the body responds to the drug.
- ❖ the study of substances that interact with living systems through chemical processes
- ❖ These substances may be
 - ✓ chemicals administered to achieve a beneficial therapeutic effect on some process within the patient or
 - ✓ their toxic effects on regulatory processes in parasites infecting the patient

Basic areas of pharmacology

- **pharmacokinetics:** deals with absorption, distribution, metabolism and excretion of drugs
- **pharmacodynamics:** the study of biochemical and physiological effects of drugs and their MOA
- **Pharmacotherapy:** the science of substances used to prevent, diagnose, and treat disease
- **Toxicology** is the branch of pharmacology that deals with the undesirable effects of chemicals on living systems.
- **Pharmacogenomics:** relationship between individual genetic makeup and his/her response to specific drugs

- ❖ The study of pharmacology encompasses
 - the sources,
 - chemical properties,
 - biological effects and therapeutic uses of drugs.

Historical perspectives

❖ In the last 1500 years

- ✓ Attempt was done to explain all of **biology and disease without the need for experimentation and observation**

Assumption = disease was caused by **excesses of bile or blood in the body**

Around the end of the 17th century

- ✓ **observation and experimentation** began to replace theorizing in **physiology and clinical medicine**
- ✓ **materia medica** was developed

Historical perspectives

- ❖ **In the late 18th and early 19th centuries**
- ✓ **François Magendie** and his student **Claude Bernard** began to develop the **methods of experimental physiology and pharmacology**
- ✓ Advances in chemistry and the further development of physiology in the 18th, 19th, and early 20th centuries laid the foundation needed for understanding **how drugs work at the organ and tissue levels.**
- ❖ **1940s and 1950s**
- ✓ Information accumulated about drug action and the **biologic substrate of that action, the drug receptor**

Drug, definitions and source

Drug

- ❖ Is any substance or product used or intended to be used to **modify** or explore the physiological state or **pathological condition** for the well being of the recipient (WHO)
- ❖ Any chemical agent that affects living protoplasm
- ❖ any substance (other than food that provides nutritional support) that, when
 - ✓ inhaled,
 - ✓ injected,
 - ✓ smoked, consumed
 - ✓ absorbed via a patch on the skin, or dissolved under the tongue causes a temporary **physiological** change in the body.

Sources

❖ 1. natural drugs

❖ **Plants;** Roots, bark, sap, leaves, flowers, seeds were sources for drugs e.g.

- Reserpine from Rauwolfia Vomitoria,

- Digitalis from foxglove,

- opium from the poppy plant.

❖ **Animal sources;** gave us hormones for replacement in times of deficiencies e.g. Insulin from the **pancreases of pigs** and cattle, Liver extracts for anemia etc

❖ **Minerals:** Iodine, iron, potassium salts.

❖ **Micro-organisms:** penicillin,

2. Synthetic drugs; from natural products in the laboratory.

3. Semisynthetic drugs: by chemical modifications of natural products

4. Biosynthetic drugs: by rDNA technology.

Drug naming system

Proprietary medicines or brand medicines

- Protected by a patent,
the patent usually covers **the formula** and manufacturing process and is usually in force for several years.
- During this time the medicines can not copied legally
- Proprietary medicines are also sold as **brand medicines**, original medicines, innovator products
- **Once the patent expires** the protected information will be part of the public domain i.e the **information** can be used by everybody.
- At this point the medicines may be used as **models** for manufacturing of generic products

International non proprietary name

- Often simply called generics. These are medicines that are copies of proprietary medicines, they contain the **same active ingredient**, at the **same dosage**, and have the **same pharmaceuticals form**.
- They can be manufactured and sold only once the **patents protecting** the proprietary medicines expired.
- Generic medicines are **cheaper** than proprietary medicines.

Chemical Name/Scientific name

-long name, refers to the chemical structure of the drug

N-acetyl-p-aminophenol For PCM

Drug categories...

a) Prescription Drugs

–Prescription drugs are drugs that the federal government has designated to be **potentially harmful** unless their use is supervised by a licensed health care provider, such as a nurse practitioner, physician, or dentist

Drug categories...

b) Nonprescription Drugs

- Drugs that are designated by the federal government to be safe (**if taken as directed**) & obtained without a prescription
- They are also referred to as **over-the-counter (OTC) drugs** & may be purchased in a variety of settings, such as a pharmacy, drugstore, etc
- They include those given for symptoms of the
 - common cold,
 - headaches,
 - constipation,
 - diarrhea, &
 - stomach upset

Drug categories...

c) Controlled Substances

- They are the most carefully monitored of all drugs
- These drugs have a **high potential for abuse** & may cause physical or psychological dependence

D. Pregnancy categories

- Medications are tested to determine the medication's effect on a fetus
- Based on these test results, medications are categorized according to their safety for being administered to pts during pregnancy

Pregnancy categories...

- **Category A:**
 - Adequate & well-controlled studies indicate **no risk** to the fetus in the first trimester of pregnancy or later
 - **levothyroxine, folic acid,**
- **Category B:**
 - Animal reproduction studies indicate **no risk to the fetus**; however, there are no well-controlled studies in pregnant women
 - Example: **metformin, hydrochlorothiazide, , amoxicillin**

Pregnancy categories...

- **Category C:**
 - Animal reproduction studies have reported adverse effects on the fetus; however,
 - there are no well-controlled studies in humans, **but potential benefits may indicate use of the drug in pregnant women despite potential risks**
 - **Example :** Fluconazole, Metoprolol, Sertraline

Pregnancy categories...

- **Category D:**
 - Positive human fetal risk has been reported from investigational or marketing experience, or human studies
 - Considering potential benefit versus risk may, in selected cases, warrant the use of these drugs in pregnant women
 - E.g **Phenytoin, Lisinopril**

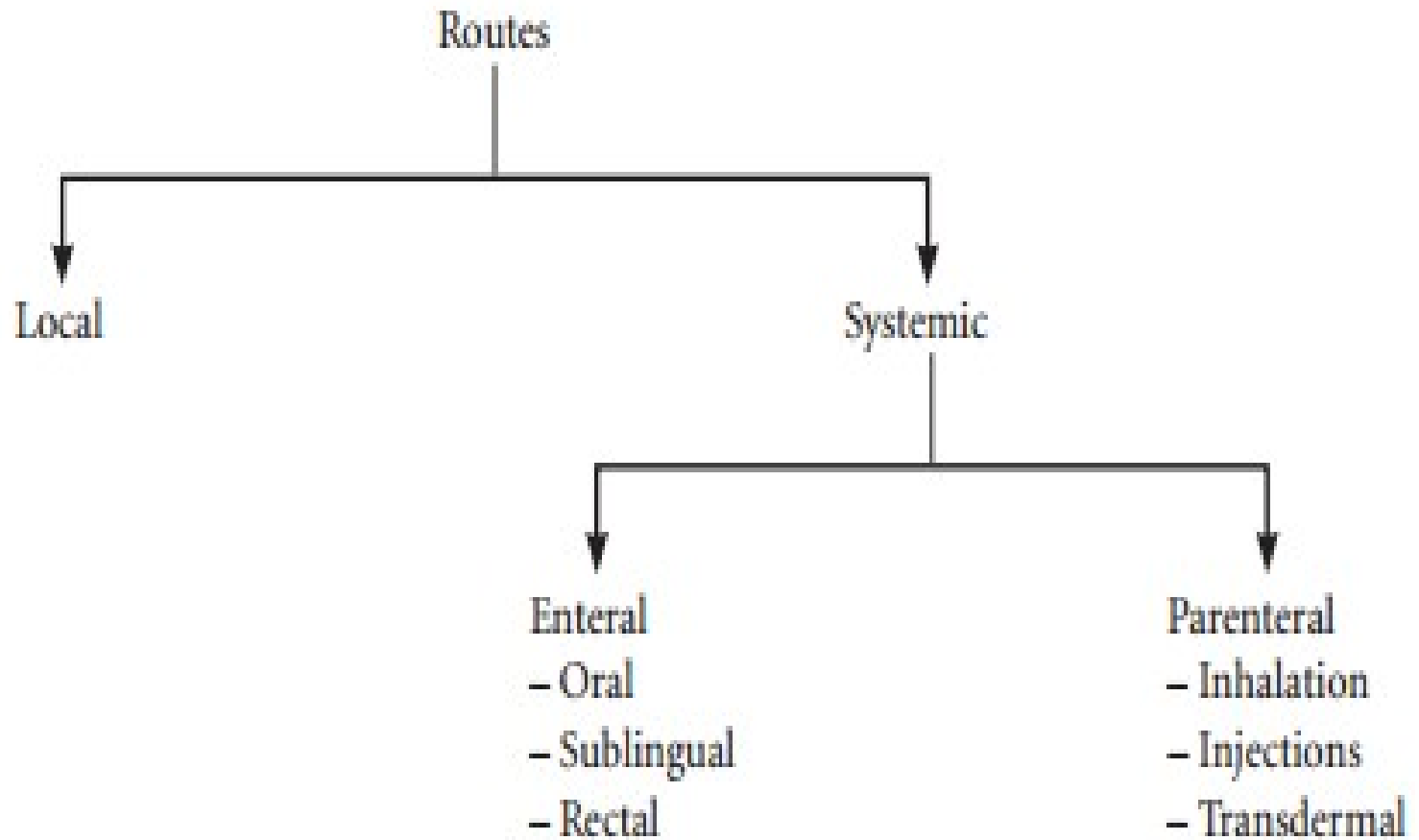
Pregnancy categories...

- **Category X:**

- Fetal abnormalities reported & positive evidence of fetal risk in humans is available from animal &/or human studies.
- The risks involved clearly outweigh the potential benefits
 - These drugs should not be administered to pregnant women

E.g Methotrexate, Warfarin

Routes of drug administration



Local Routes

- ❖ It is the simplest mode of administration of a drug at the site where the desired action is required.
- ❖ Systemic side **effects are minimal**.

Local routes can be delivered by the following

- ❖ **Oral cavity:** As a suspension, e.g. nystatin; as or as oral Jel. clotrimazole (for oral candidiasis);
- ❖ **GI tract:** As tablet that is not absorbed, e.g. neomycin (for sterilization of gut before surgery)

Local Routes

- ❖ **Eye, ear and nose:** As drops, ointments and sprays (for infection, allergic conditions, etc.), e.g. gentamicin eye/ear drops.
- ❖ **Bronchi:** As inhalation, e.g. salbutamol, ipratropium bromide, etc. for bronchial asthma
- ❖ **Skin:** As ointment, cream, lotion or powder, e.g. clotrimazole (antifungal) for cutaneous candidiasis.

Enteral Routes

- ❖ It includes oral, sublingual and rectal routes
- ❖ **Oral Route**
 - ✓ It is the most common and acceptable route for drug administration.
 - ✓ Dosage forms are tablet, capsule, syrup, mixture, etc., e.g., **paracetamol tablet for fever, omeprazole capsule for peptic ulcer** are given orally

Advantages

- Safer. cheaper, painless. convenient for repeated and prolonged use. can be self-administered.

Dis advantage

- Not suitable for **emergency** as onset of action of orally administered drugs is slow.

❖ Sublingual Route

- The preparation is kept under the tongue.
- The drug is absorbed through the buccal mucous membrane and enters the systemic circulation directly, e.g. **nitroglycerin** for **acute anginal** attack and **buprenorphine** for **myocardial infarction**
- **Advantages**
 - Quick onset of action.
 - Action can be terminated by spitting out the tablet.
 - Bypasses first-pass metabolism.
 - Self-administration is possible.

❖ **Sublingual Route...**

➤ **Disadvantage...**It is not suitable for:

- Irritant and lipid-insoluble drugs.
- Drugs with bad smell and taste.

❖ **Rectal Route**

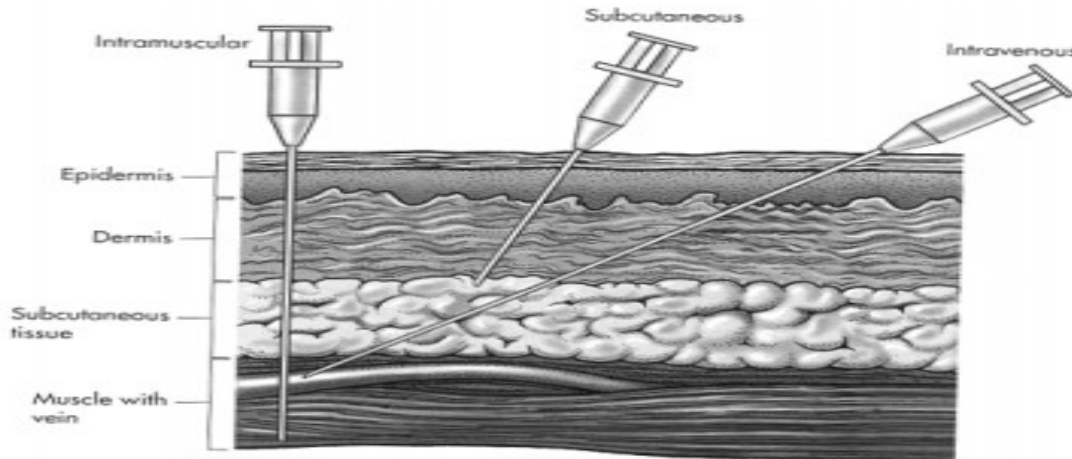
Drugs can be given in the form of solid or liquid.

- **Suppository:** It can be used for local (topical) effect (as well as systemic effect, e.g. **indomethacin** for **rheumatoid arthritis**).
- **Enema:** Retention enema can be used for local effect as well as systemic effect.

Parenteral Routes

- Routes of administration other than enteral route are called **parenteral routes**
- **Advantages**
 - ✓ Onset of action of drugs is faster; hence it is suitable for emergency.
 - ✓ Can be for Unconscious patient.
 - ✓ Uncooperative patients.
 - ✓ Patients with vomiting and diarrhea.
- **Disadvantages**
 - ✓ Require aseptic conditions.
 - ✓ Preparations should be sterile and is expensive.
 - ✓ Requires invasive techniques that are painful.
 - ✓ Cannot be usually self-administered.
 - ✓ Can cause local tissue injury to nerves, vessels, etc.

Injection sites



Subcutaneous (s.c.) route

- The drug is injected into the subcutaneous tissues of the thigh, abdomen and arm, e.g. adrenaline, insulin, etc
- **Advantages**
 - ✓ Self-administration is possible (e.g. insulin).
 - ✓ Depot preparations can be inserted into the subcutaneous tissue, e.g. norplant for contraception.

Disadvantages

- ✓ It is suitable only for nonirritant drugs.
- ✓ Drug absorption is slow; hence it is not suitable for emergency.

Intramuscular (i.m.) route: Drugs are injected into large muscles such as deltoid, gluteus maximus and vastus lateralis,

e.g., diclofenac, etc. A volume of 5–10 mL can be given at a time.

Advantages

- ✓ Absorption is more rapid as compared to oral route.
- ✓ Mild irritants, depot injections, soluble substances and suspensions can be given by this route.

Disadvantages

- ✓ Aseptic conditions are needed.
- ✓ Intramuscular injections are painful and may cause abscess.
- ✓ Self-administration is not possible.
- ✓ There may be injury to the nerves.

Intravenous (i.v.) route: Drugs are injected directly into the blood stream through a vein.

Drugs can be given as

Bolus: Single, relatively large dose of a drug injected rapidly or slowly as a single unit into a vein. For example, i.v. ranitidine in bleeding peptic ulcer

Slow Intravenous infusion: For example, dopamine infusion in cardiogenic shock

Advantage

- ✓ Bioavailability is 100%
- ✓ Quick onset of action
- ✓ Large volume of fluid can be administered

✓ **Disadvantage**

- ✓ Once the drug is injected, its action cannot be halted.
- ✓ Self-medication is not possible.
- ✓ Strict aseptic conditions are needed.
- ✓ Extravasation of some drugs can cause injury, necrosis and sloughing of tissues.

Mechanisms of drug permeation

- ❖ Drug permeation proceeds by several mechanisms.
- ✓ **Passive diffusion** in an aqueous or lipid medium is common,
- ✓ **active processes** play a role in the movement of many drugs, especially those whose molecules are too large to diffuse readily

Aqueous diffusion: Aqueous diffusion of drug molecules is usually driven by the concentration gradient of the permeating drug

- permit the passage of molecules as large as MW 20,000–30,000 Da

Mechanisms of drug permeation...

- ❖ Factors affecting diffusion
 - ❖ bound to large plasma protein
 - ❖ Charging of the drug
- ❖ The passive flux of molecules down a concentration gradient is given by Fick's law

Flux (molecules per unit time) =

$$(C_1 - C_2) \times \frac{\text{Area} \times \text{Permeability coefficient}}{\text{Thickness}}$$

Active transport

Active transport

- ✓ is the movement of molecules across a membrane from a region of lower concentration to a region of higher concentration—against the concentration gradient.
- ✓ Active transport requires cellular energy to achieve this movement

❖ *Special carriers* _ Special carrier molecules exist for many substances that are important for cell function and too large or *too insoluble* in lipid to diffuse passively through membranes, eg, **peptides, amino acids, and glucose**

Special carriers...

- ❖ movement is through by *active transport* or **facilitated diffusion**
- ❖ They are selective and **saturable**

Endocytosis and exocytosis _ A few substances are so **large** or **impermeant** that they can enter cells only by endocytosis,

- ✓ In endocytosis the substance is bound at a **cell surface** receptor, **engulfed by the cell membrane**, and carried into the cell by pinching off of the newly formed vesicle inside the membrane.
- ✓ The substance can then be released inside the cytosol by breakdown of the vesicle membrane

E.g Vit B complex and iron

- ✓ Exocytosis is the reverse process
- ✓ for the secretion of many substances from cells

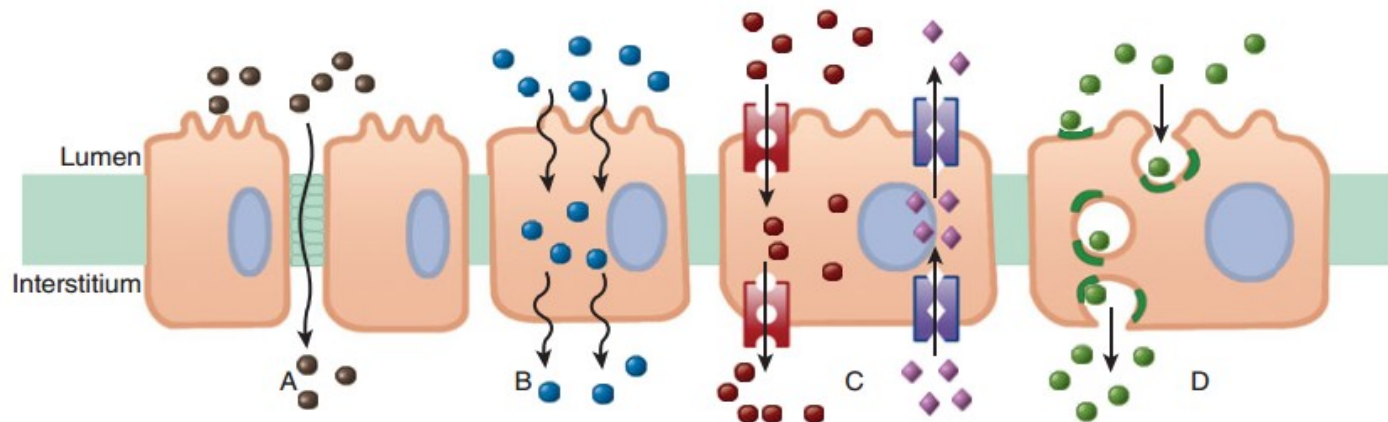


fig : mechanisms of drug permeation

Keys: a: passive diffusion

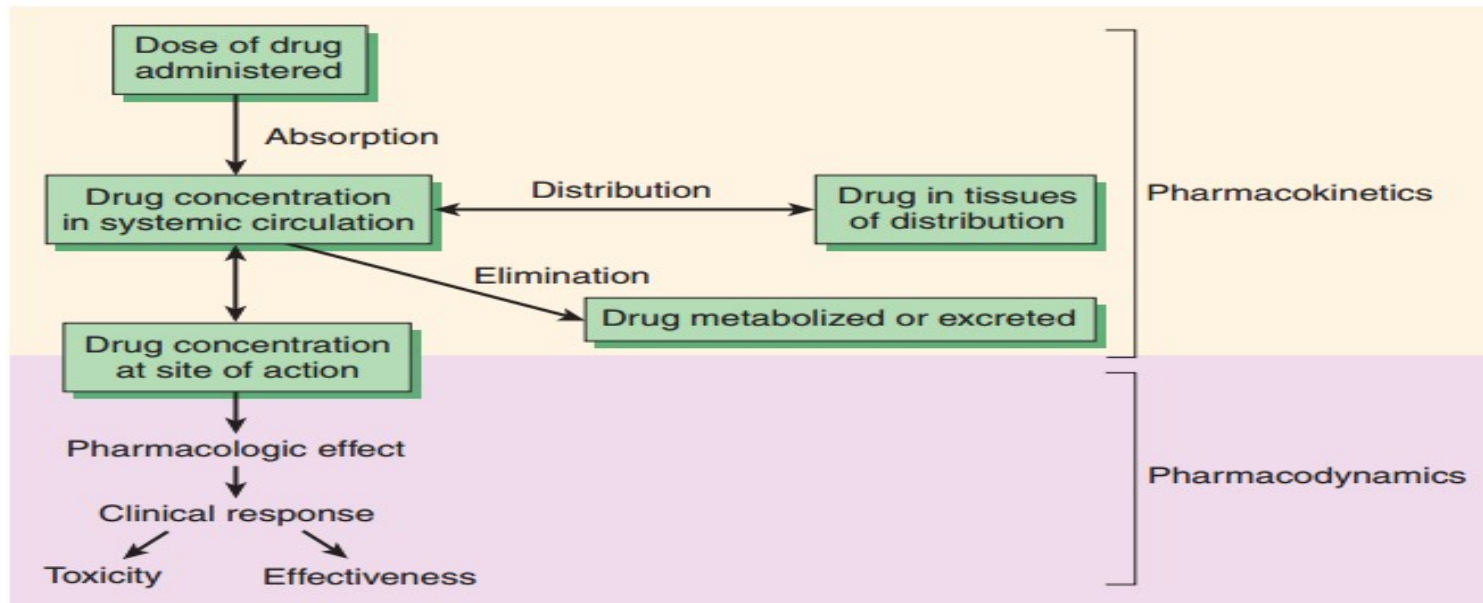
c: endocytosis

b: Facilitated diffusion

d: exocytosis

Pharmacokinetics

- Pharmacokinetics (PK) deals with
- drug absorption, distribution, metabolism and excretion (ADME)



Absorption

- ❖ **Absorption** is the movement of a drug from its site of administration into the **central compartment**.
- ❖ For **solid dosage** forms, *absorption* first requires dissolution of the tablet or capsule, thus liberating the drug
- ❖ **Bioavailability** describes the fractional extent to which a dose of drug reaches its site of action or a biological fluid.
- ❖ **Factors affecting drug absorption**
 - Patient related factors
 - Pharmaceutical factors

Patient related factors

❖ Patient factors is related to the following

➤ Gastric emptying rate

Drug swallowed-----reaches to stomach.....small intestine (duodenum), if a delay in release of gastric content to small intestine.....***absorption delays....irritation/ instability will happen***

✓ Gastric emptying rate is faster in case of **solution & suspensions** than solid & non disintegrating dosage forms

➤ Intestinal motility

➤ Normal peristaltic movements..... drug will have a sufficient time (*residence time*) at the absorption site for optimum absorption.

➤ **Blood flow to GIT...greater blood flow greater absorption**

Patient related factors...

- ❖ Effect of Food... it may increase or decrease
- ❖ penicillin and tetracycline..... ↓ griseofulvin↑

Pharmaceutical factors

Pharmaceutical factors is related to

- drug solubility & dissolution rate
- Particles size & effective surface area
- Solvates & hydrates
- salt form of drug
- Ionization state

Distribution of Drug

❖ ***Distribution*** is a movement of drug to site of ***action***

- a drug distributes into **interstitial** and **intracellular fluids** as functions of the physicochemical properties of the drug,
- the rate of drug delivery is at different capacities at different regions to interact with the drug.
 - Cardiac output,
 - regional blood flow,

Distribution of Drug...

❖ well-perfused organs

- liver,
- kidney,
- Brain.

❖ Less perfused organs

- muscle,
- most viscera, skin, and
- fat

VOLUME OF DISTRIBUTION

- Apparent volume of distribution (V_d)

Presuming that the body behaves as a single homogeneous compartment with volume V into which drug gets immediately and uniformly distributed

$$V = \frac{\text{dose administered i.v.}}{\text{plasma concentration}}$$

Distribution of Drug...

Binding to Plasma Proteins.... Many drugs circulate in the bloodstream bound to plasma proteins.

- Albumin is a major carrier for acidic drugs;
- α 1-acid glycoprotein binds basic drugs.
- Nonspecific binding to other plasma proteins generally occurs to a much smaller extent.
- The binding is usually **reversible**.

Distribution of Drug...

Tissue Binding

- ❖ Many drugs accumulate **in tissues** at higher concentrations than those in the extracellular fluids and blood.
- ❖ Will serve as a **reservoir** that prolongs drug action in
- ❖ accumulation also can produce **local toxicity** (e.g., **renal** and **ototoxicity** associated with aminoglycoside antibiotics).

Distribution of Drug...

CNS, the BBB, and CSF

- The brain capillary endothelial cells have continuous **tight junctions**; therefore, drug penetration into the brain depends on **transcellular** rather than **paracellular** transport.
- The **lipid solubility** of the nonionized and unbound species of a drug is an important determinant of its uptake by the brain;
- The more lipophilic a drug, the more likely it is to cross the BBB.

Distribution of Drug...

Bone

- **Bone** is a reservoir for the slow release of toxic agents such as **lead** or **radium**;
- their effects thus can persist long after exposure.
- **Local destruction** of the bone medulla also may result in reduced blood flow

Distribution of Drug...

Fat

- Fat will serve as a **reservoir**
- **Many lipid-soluble** drugs are stored by physical solution in the neutral fat.
- In obese persons, the fat content of the body may be as high as **50%**, and even in lean individuals, fat constitutes **10%** of body weight; hence, fat may serve as a reservoir for lipid-soluble drugs.
- Fat is a rather **stable reservoir** because it has a relatively low blood flow.

Metabolism

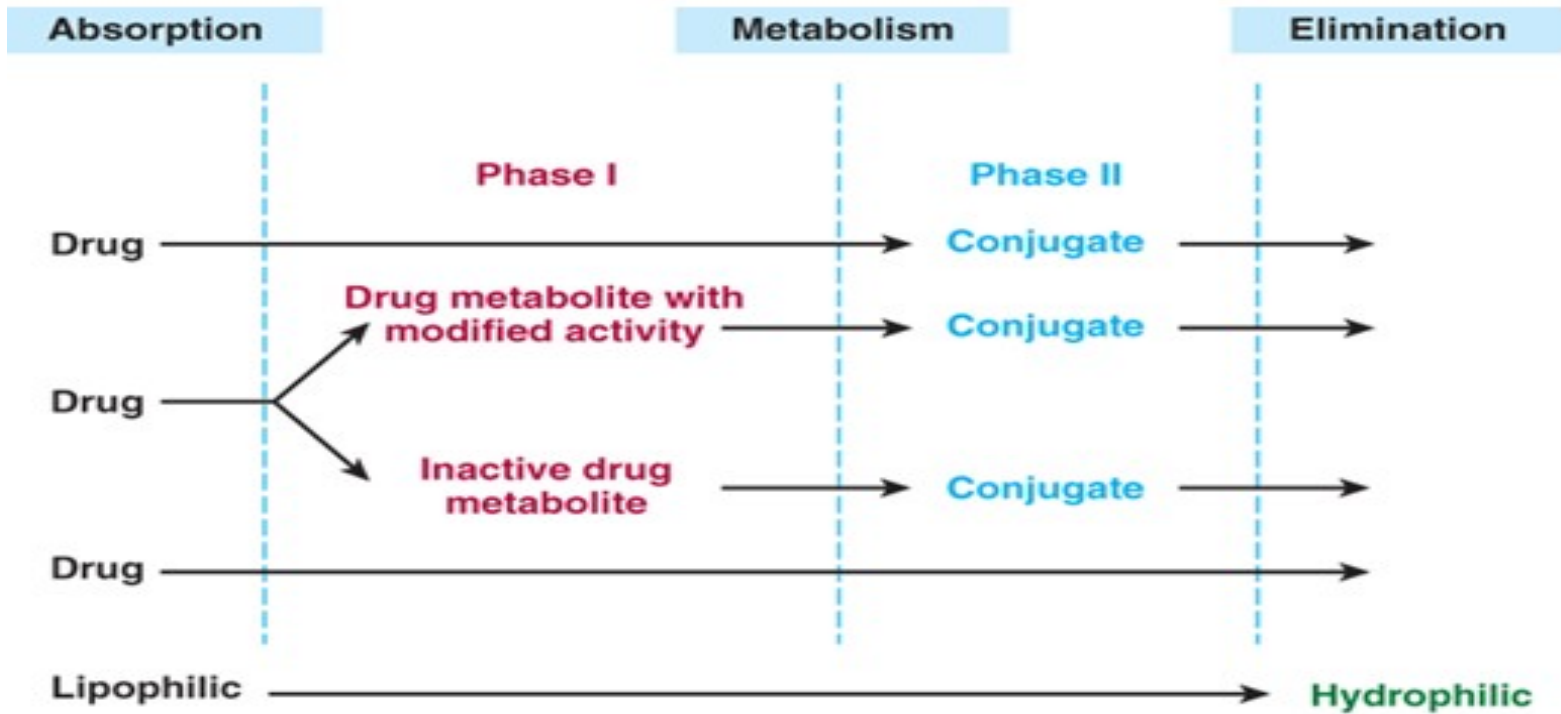
❖ The metabolism of drugs and other xenobiotics into more hydrophilic metabolites is

- essential for the elimination of these compounds from the body and termination of their biological activity.

First-Pass Metabolism

- Following **non parenteral** administration of a drug,
- a significant portion of the dose may be metabolically inactivated in either the intestinal **endothelium** or the **liver** before it reaches the **systemic circulation**
- Limits oral availability of highly **metabolized drugs**

Metabolism



Biotransformation

- ✓ Generates more polar (water soluble), inactive metabolites
- ✓ Readily **excreted from body**
- ✓ **Metabolites** may still have potent biological activity (or may have toxic properties)
- ✓ Generally applicable to metabolism of all **xenobiotics** as well as **endogenous compounds**

Phase I and Phase II Metabolism

- Phase I
 - functionalization reactions
- Phase II
 - conjugation reactions

Phase I

- ❖ Converts the parent drug to a more polar metabolite by introducing or unmasking a functional group (-OH, -NH₂, -SH).
- ❖ Usually results in **loss** of pharmacological activity
- ❖ **Sometimes may** be equally or more active than parent

Phase I...

- ✓ Includes **oxidation, reduction, hydrolysis**, and hydration and isomerization
- ✓ Many drugs undergo a **number of these reactions**
- ✓ Main function of Phase I metabolism is to prepare the compound for phase **II metabolism**
- ✓ Mixed function enzyme system found in microsomes of many cells (esp **liver, kidney, lung, intestine**) performs many different functionalization reactions

Phase II Metabolism

- ❖ Phase II is usually the true detoxification of drugs
- ❖ Occurs mostly in cytosol
- ❖ Gives products that are generally water soluble and easily excreted

Includes sugar conjugation, sulfation, methylation, acetylation, amino acid conjugation, glutathione conjugation

Phase II (conjugation reactions)...

- ✓ Subsequent reaction in which a covalent linkage is formed between a functional group on the parent compound or Phase I metabolite and an endogenous substrate such as *glucuronic acid*, **sulfate**, **acetate**, or an **amino acid**
- ✓ **Highly polar** – rapidly excreted in **urine and feces**
- ✓ **Usually inactive** - notable exception is **morphine**

Site of Biotransformation

- ✓ Enzymatic in nature
- ✓ Enzyme systems involved are localized in **liver**
- ✓ Every tissue has some metabolic activity
- ✓ Other organs with significant metabolic capacity are **GI tract, kidneys and lung**

Excretions of drugs

- ❖ Drugs are eliminated from the body either **unchanged** or as **metabolites**.
- ❖ Excretory organs, eliminate **polar compounds** more efficiently than substances with high lipid solubility.
- ❖ The **kidney** is the most important organ for excreting drugs and their metabolites.
- ❖ Renal excretion of unchanged drug is a major route of elimination for **25-30%** of drugs administered to humans..

Excretions of drugs

- ❖ **breast milk** is important not because of the amounts eliminated, but because the excreted drugs may affect the nursing infant.
- ❖ Excretion from the **lung** is important mainly for the elimination of **anesthetic gases**

Excretions of drugs

Renal excretion

- ❖ Excretion of drugs and metabolites in the urine involves 3 distinct processes: **glomerular filtration**, **active tubular secretion**, and **passive tubular reabsorption**.
- ❖ Changes in overall renal function generally affect all 3 processes to a similar extent.
- ❖ In **neonates**, **renal function** is low compared with body mass but matures rapidly within the first few months after birth.
- ❖ During adulthood, there is a slow decline in renal function, $\sim 1\%$ per year, so that in elderly patients a **substantial degree of functional impairment** may be present.

Kinetics Of Elimination

- Rate of elimination is the amount of drug eliminated (in units of weight like grams) per unit time.
- If it is seen as a function of plasma concentration, we derive an important parameter known as clearance (**CL**).
- Thus **clearance** is the rate of elimination of a drug divided by its plasma concentration.

First Order kinetics (Linear kinetics)

1. Constant fraction of drug is eliminated per unit time.
2. Rate of elimination is proportional to plasma concentration.
3. Clearance remains constant.
4. Half life remain constant.
5. Most of the drugs follow first order kinetics.

Zero Order kinetics (Non linear kinetics)

1. Constant amount of the drug is eliminated per unit time.
2. Rate of elimination is independant of plasma concentration.
3. Clearance is more at low concentrations and less at high conc.
4. Half life is less at low conc. and more at high conc.
5. Very few drugs follow pure zero order kinetics e.g. alcohol.
6. Any drug at high conc. (when metabolic or elimination pathway is saturated) May show zero order kinetics.

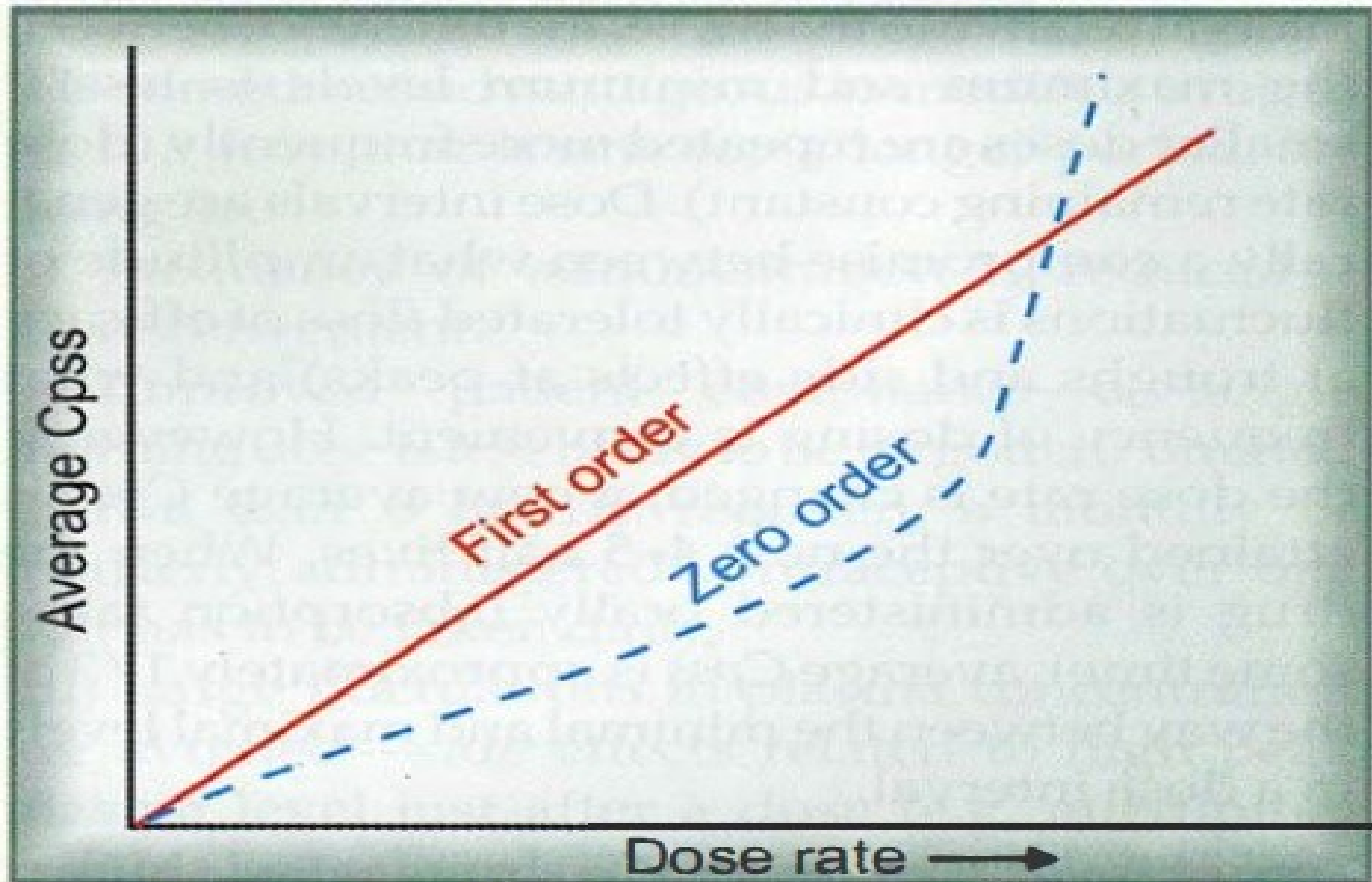


Fig. 3.5: Relationship between dose rate and average steady-state plasma concentration of drugs eliminated by first order and Michaelis Menten (zero order) kinetics

Plasma half-life

1. The Plasma half-life ($t_{1/2}$) of a drug is the time taken for its plasma concentration to be reduced to half of its original value.
 - initial rapidly declining (α) phase-due to distribution.
 - later less declined (β) phase-due to elimination.

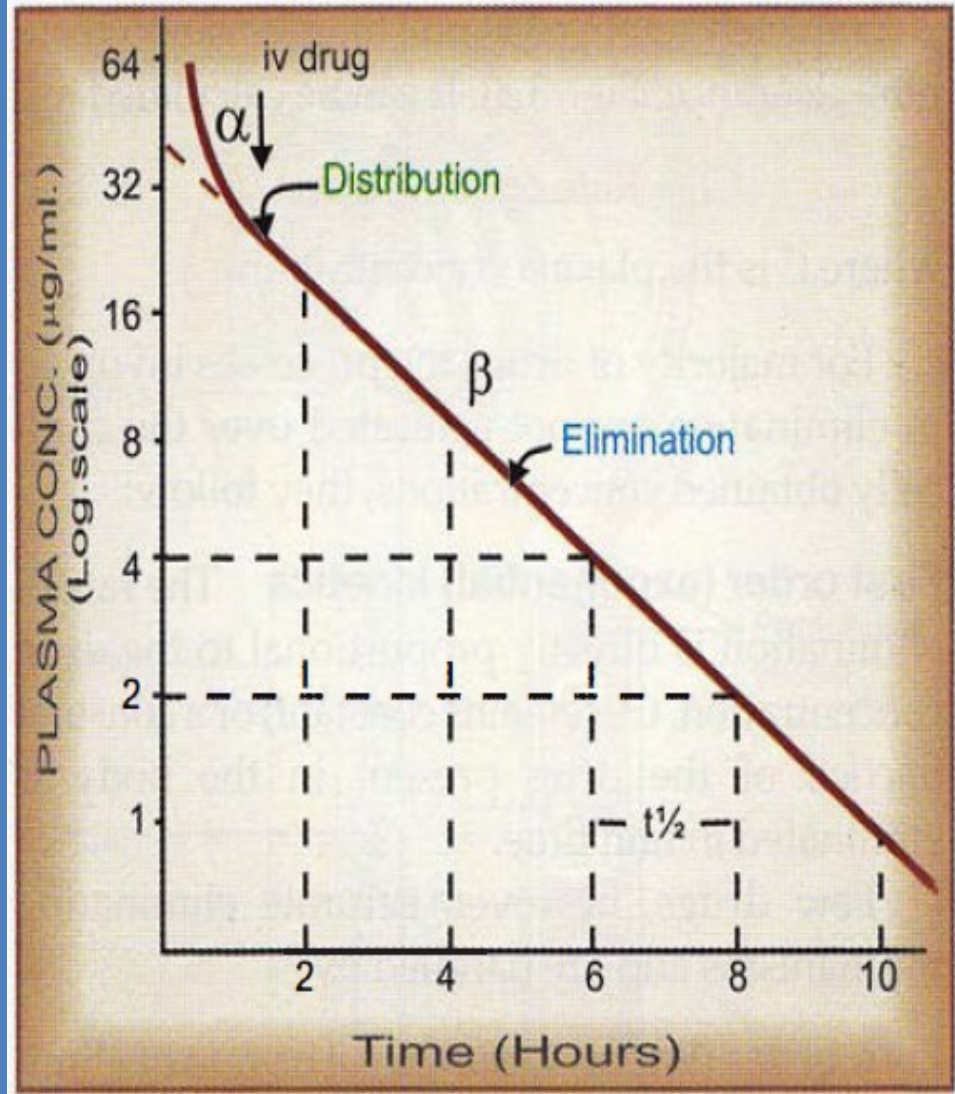


Fig. 3.4: Semilog plasma concentration-time plot of a drug eliminated by first order kinetics after intravenous injection